Using the actual time points

$$t = (6, 9, 12, 15, 18),$$

with

$$\mathbf{G} = \begin{pmatrix} 9.6144 & -0.47 \\ -0.47 & 0.031 \end{pmatrix}$$

and residual variance

$$\sigma_e^2 = 9.0241,$$

we can calculate the marginal covariance and correlation matrices directly.

For a random-intercept/random-slope model,

$$\mathbf{y}_i = \mathbf{X}_i \beta + \mathbf{Z}_i \mathbf{b}_i + \epsilon_i,$$

with

$$\mathbf{Z}_i = \begin{pmatrix} 1 & 6 \\ 1 & 9 \\ 1 & 12 \\ 1 & 15 \\ 1 & 18 \end{pmatrix},$$

the marginal covariance matrix is

$$\mathbf{V} = \mathbf{Z}_i \mathbf{G} \mathbf{Z}_i^{\mathsf{T}} + \sigma_e^2 \mathbf{I}.$$

Marginal covariance matrix

This gives

$$\boxed{\mathbf{V} = \begin{pmatrix} 14.1145 & 4.2384 & 3.3864 & 2.5344 & 1.6824 \\ 4.2384 & 12.6895 & 3.0924 & 2.5194 & 1.9464 \\ 3.3864 & 3.0924 & 11.8225 & 2.5044 & 2.2104 \\ 2.5344 & 2.5194 & 2.5044 & 11.5135 & 2.4744 \\ 1.6824 & 1.9464 & 2.2104 & 2.4744 & 11.7625 \end{pmatrix}}$$

Notice that the marginal variances are **not constant** because the random-slope term makes them depend on time.

Specifically,

$$\mathrm{Var}(Y_t) = 18.6385 - 0.94t + 0.031t^2.$$

Thus the variances at $t = 6, 9, 12, 15, 18$ are

$$(14.1145,\ 12.6895,\ 11.8225,\ 11.5135,\ 11.7625).$$

Marginal correlation matrix

Standardizing $\mathbf{V}$ gives

$$\boxed{\mathbf{C} = \begin{pmatrix} 1.000 & 0.317 & 0.262 & 0.199 & 0.131 \\ 0.317 & 1.000 & 0.252 & 0.208 & 0.159 \\ 0.262 & 0.252 & 1.000 & 0.215 & 0.187 \\ 0.199 & 0.208 & 0.215 & 1.000 & 0.213 \\ 0.131 & 0.159 & 0.187 & 0.213 & 1.000 \end{pmatrix}}$$

So, for example:

$$\rho(Y_6, Y_9) = 0.317,$$

$$\rho(Y_6, Y_{18}) = 0.131,$$

and

$$\rho(Y_{15}, Y_{18}) = 0.213.$$

An important feature of these results

With these particular estimates, the marginal correlation **does not simply decrease with increasing time separation**. For example,

$$\rho(6, 18) = 0.131$$

is smaller than

$$\rho(15, 18) = 0.213,$$

but the intermediate correlations reflect both the negative intercept–slope covariance and the changing marginal variances.

The general pairwise formula you're using is

$$\boxed{\mathrm{Cov}(Y_{t_j}, Y_{t_k}) = 9.6144 - 0.47(t_j + t_k) + 0.031 t_j t_k}$$

and

$$\boxed{\rho(t_j, t_k) = \frac{\mathrm{Cov}(Y_{t_j}, Y_{t_k})}{\sqrt{\mathrm{Var}(Y_{t_j})\,\mathrm{Var}(Y_{t_k})}}.}$$

One potential issue worth checking is **how time was coded in the actual model**. If the model used centered time (e.g., $t - 12$) or scaled time rather than the raw values $6, 9, 12, 15, 18$, the calculation must use that transformed time in $\mathbf{Z}$. This can substantially change the resulting correlation matrix.



Cloudflare Ad

100+ Hands-On Workshops

Watch live code demos and talk architecture with our core engineers.